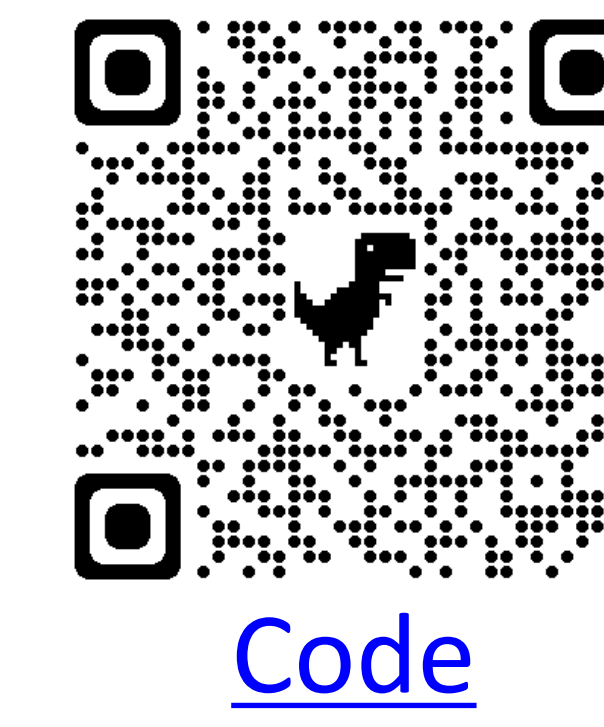




Introduction

- Large-scaled pre-trained language models suffer from various adversarial attacks

Why?

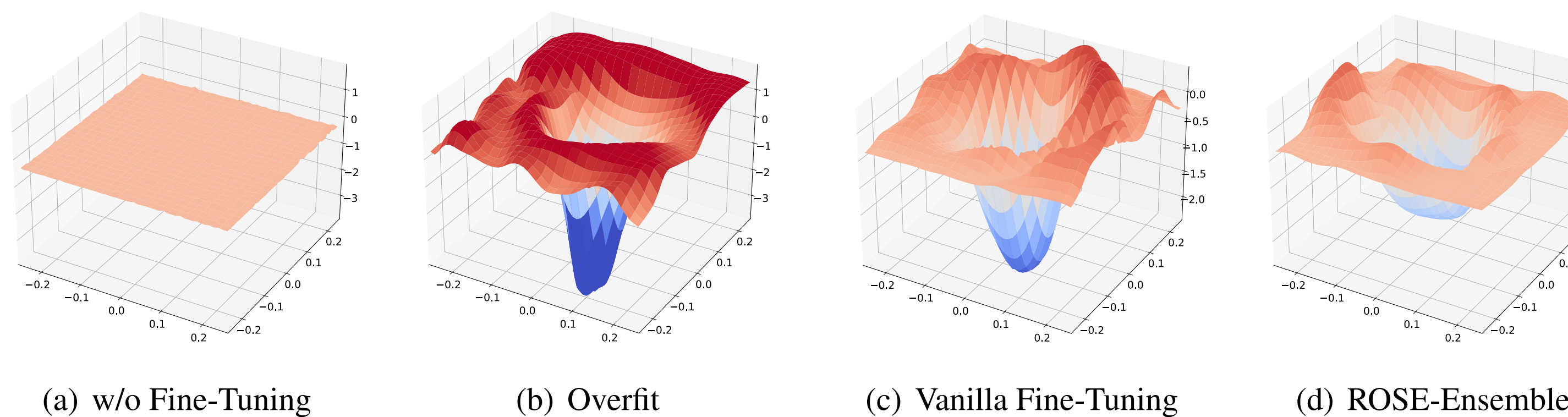
- Tend to capture spurious patterns when fine-tuning
 - Fail to utilize general linguistic knowledge learned from the pre-training stage
 - Existing defense methods: covering more diverse data
- Limited to practice!**
- Often require prohibitively large attack search spaces
 - Not generally applicable to different types of attacks

- Robust Selective Fine-tuning (ROSE)

Attack-agnostic and **Model-agnostic**

Intuition: only robust and informative updates of parameters should be conducted

- First-order ROSE: counter superficial patterns among examples with similar semantics in the current step (local)
 - Second-order ROSE: counter superficial patterns across examples with different semantics along the fine-tuning process (global)
 - ROSE-Ensemble: aggregate benefits of the above two strategies
- ROSE leads to solutions with **broader and flatter** optima, which corresponding to better adversarial robustness:

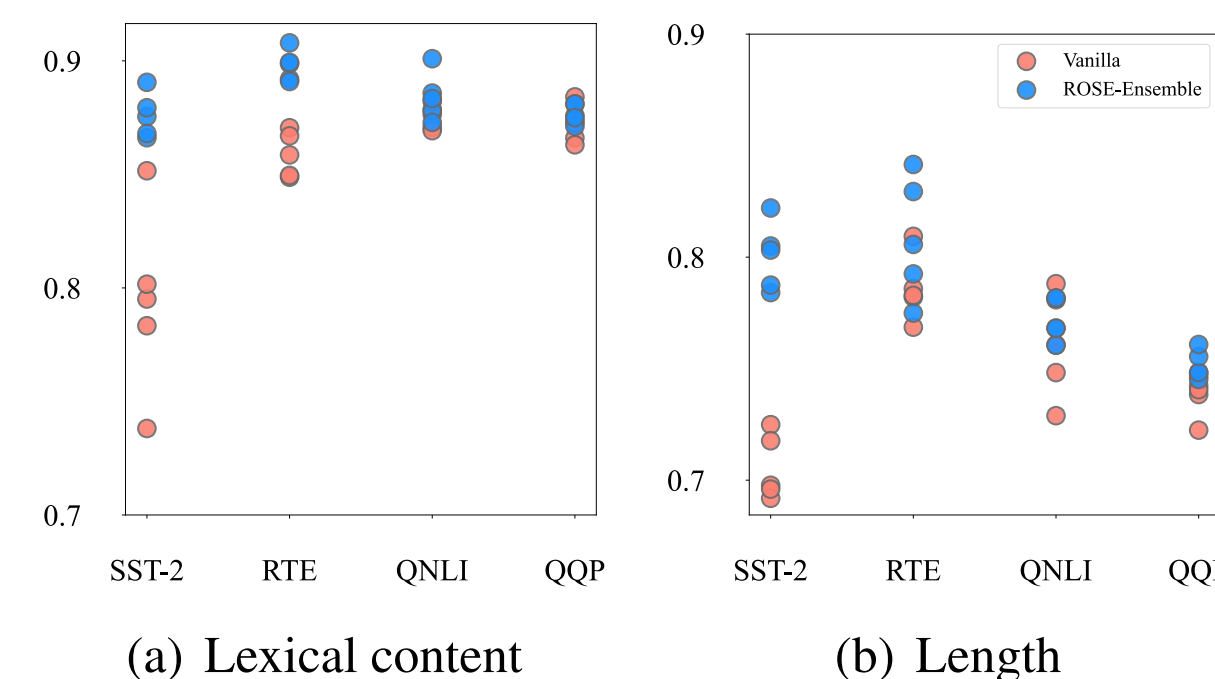


Experiments

- Effect of γ

γ	RoBERTa _{BASE}		RoBERTa _{LARGE}	
	Avg	$\Delta \downarrow$	Avg	$\Delta \downarrow$
	GLUE+AdvGLUE		GLUE+AdvGLUE	
0.1	59.35	56.81	75.07	28.99
0.3	58.42	55.13	75.51	30.27
0.5	65.02	47.84	78.10	27.67
0.7	61.09	51.36	75.60	30.52
0.9	59.88	50.82	73.22	32.25

- Preference for deep features



Method

ROSE-First

- Employ dropout as adversarial perturbations:
 $\mathcal{L}_t^{\text{KL}} = \mathcal{D}_{\text{KL}}(P_t \parallel P'_t) + \mathcal{D}_{\text{KL}}(P'_t \parallel P_t)$.
- Determine potential risk to capture superficial cues of the i -th parameter:
 $r_t^{\text{F},i} = \|\nabla \mathcal{L}_t^{\text{KL}}\|_i(\theta_{t-1})\|_F$.
- Given the upper threshold for robust parameters, the mask can be derived as:

$$\mathbf{M}_t^{\text{F},k_i} = \begin{cases} 1 & i/n \leq c_h, \\ 0 & \text{otherwise.} \end{cases}$$

ROSE-Second

ROSE-Second

$$r_t^{\text{S},i} = \left| \frac{\alpha \|\mathbf{g}_t^i\|_F}{\|\mathbf{m}_{t-1}^i\|_F} - 1 \right| \quad \mathbf{M}_t^{\text{E}} = \gamma \mathbf{M}_t^{\text{F}} + (1 - \gamma) \mathbf{M}_t^{\text{S}}$$

Algorithm 1 ROSE for AdamW Optimizer

given $\alpha = 0.001, \beta_1, \beta_2 \in [0, 1), \epsilon = 10^{-8}, \lambda \in \mathbb{R}$
1: **initialize** time step $t \leftarrow 0$, parameter vector $\theta_{t=0} \in \mathbb{R}^n$, first moment vector $\mathbf{m}_{t=0} \leftarrow \mathbf{0}$, second moment vector $\mathbf{v}_{t=0} \leftarrow \mathbf{0}$, learning rate $\eta \in \mathbb{R}$
2: **repeat**
3: $t \leftarrow t + 1$
4: $\mathbf{g}_t \leftarrow \nabla \mathcal{L}_t^{\text{SCE}}(\theta_{t-1})$
5: $\mathbf{r}_t^{\text{F}} \leftarrow \|\nabla \mathcal{L}_t^{\text{KL}}(\theta_{t-1})\|_F$ ▷ calculate first-order risks
6: $\mathbf{r}_t^{\text{S}} \leftarrow |(1 - \beta_1)|\mathbf{g}_t|_F / \|\mathbf{m}_{t-1}\|_F - 1|$ ▷ calculate second-order risks
7: $\mathbf{M}_t \leftarrow \text{CalculateMask}(\mathbf{r}_t^{\text{F}}, \mathbf{r}_t^{\text{S}})$ ▷ calculate mask
8: $\mathbf{g}_t' \leftarrow \mathbf{M}_t \odot \mathbf{g}_t + (\mathcal{J} - \mathbf{M}_t) \odot \mathbf{m}_{t-1}$ ▷ update gradients
9: $\mathbf{m}_t \leftarrow \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t'$
10: $\mathbf{v}_t \leftarrow \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_t'^2$
11: $\hat{\mathbf{m}}_t \leftarrow \mathbf{m}_t / (1 - \beta_1^t)$
12: $\hat{\mathbf{v}}_t \leftarrow \mathbf{v}_t / (1 - \beta_2^t)$
13: $\theta_t \leftarrow \theta_{t-1} - \eta \left(\hat{\mathbf{m}}_t / (\sqrt{\hat{\mathbf{v}}_t} + \epsilon) + \lambda \theta_{t-1} \right) \odot \mathbf{M}_t$ ▷ update weights
14: **until** stopping criterion is met
15: **return** optimized parameters θ_t

Experiments

- Model performance on GLUE and AdvGLUE benchmarks

Model	SST-2		RTE		QNLI		QQP		Avg	Avg $\Delta \downarrow$
	GLUE	AdvGLUE	GLUE	AdvGLUE	GLUE	AdvGLUE	GLUE	AdvGLUE	GLUE+AdvGLUE	
RoBERTa _{BASE}										
Vanilla	94.29	24.05	77.91	28.15	92.97	27.43	91.58	19.49	56.98	64.41
R-Drop	95.32	27.84	79.86	31.36	93.30	28.92	91.86	37.44	60.74	58.70
CHILD-TUNING _D	94.21	23.82	75.52	16.54	92.36	31.89	91.64	17.95	55.49	65.88
SMART	94.98	35.95	77.54	24.44	93.35	34.29	91.04	46.58	62.27	53.91
FreeLB	94.89	35.81	78.42	32.10	93.12	36.22	92.04	44.10	63.34	52.56
ROSE-First	94.84	37.67	78.34	35.49	92.19	44.19	89.56	44.44	64.59	48.29
ROSE-Second	93.78	36.99	78.16	37.97	92.41	34.63	90.48	45.73	63.77	49.88
ROSE-Ensemble	94.09	39.36	78.63	38.02	92.64	39.59	90.39	47.44	65.02	47.84
RoBERTa _{LARGE}										
Vanilla	96.08	56.08	85.92	61.73	94.58	63.38	92.09	40.60	73.81	36.72
R-Drop	96.59	53.38	85.56	66.67	95.01	55.95	92.35	44.80	73.79	37.18
CHILD-TUNING _D	95.91	51.35	85.92	61.73	94.30	58.11	92.03	43.59	72.87	38.35
SMART	96.67	59.12	85.02	69.14	94.91	61.04	92.12	50.85	76.11	32.14
FreeLB	96.49	59.32	86.76	66.91	94.99	62.30	92.60	48.21	75.95	33.53
ROSE-First	95.58	57.77	85.13	70.62	94.08	64.02	90.67	60.26	77.27	28.20
ROSE-Second	96.29	60.59	85.08	67.49	94.72	63.68	91.68	55.90	76.93	30.03
ROSE-Ensemble	96.10	60.81	85.92	71.11	94.26	64.64	91.46	60.51	78.10	27.67

- Extensions of ROSE to existing method R-Drop

Model	SST-2		RTE		QNLI		QQP		Avg	Avg $\Delta \downarrow$
	GLUE	AdvGLUE	GLUE	AdvGLUE	GLUE	AdvGLUE	GLUE	AdvGLUE	GLUE+AdvGLUE	
RoBERTa _{BASE}										
R-Drop	95.32	27.84	79.86	31.36	93.30	28.92	91.86	37.44	60.74	58.70
+ ROSE-First	94.77	32.30	78.34	36.05	92.92	39.46	90.62	42.82	63.41	51.51
+ ROSE-Second	95.05	30.00	73.93	36.79	92.60	32.03	91.96	39.23	61.45	53.87
+ ROSE-Ensemble	92.36	35.68	77.83	36.54	92.50	40.27	90.90	42.56	63.58	49.64
RoBERTa _{LARGE}										
R-Drop	96.59	53.38	85.56	66.67	95.01	55.95	92.35	44.80	73.79	37.18
+ ROSE-First	96.54	55.68	85.85	68.89	94.38	56.62	91.95	53.21	75.39	33.58
+ ROSE-Second	96.63	55.81	85.27	66.91	94.84	57.16	92.38	50.26	74.91	34.74
+ ROSE-Ensemble	96.72	56.62	85.20	69.13	94.81	57.97	92.27	54.61	75.92	32.66

Analysis

- Two-dimensional loss visualization

ROSE-First → wider bottoms **Local**
ROSE-Second → less dense contour **Global**
ROSE-Ensemble → both two **Both**

